Assignment 3

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Que 1 Write LaTeX code to produce the following expression:

$$\overbrace{a+b+c}^{FirstGroup}+\underbrace{x+y+z}_{SecondGroup}=\overbrace{\underbrace{a+b}_{Sum1}+c}^{Total}$$

Que 2 Create the following array using the array environment:

$$A=\left[\begin{array}{c|ccc} & b & c & d \\ \hline a & b & c & d \\ a & b & c & d \\ a & b & c & d \end{array}\right]$$

Que 3 Write the following piecewise function using the cases environment:

$$f(x)=\begin{cases} x^2+2x+1, & x<0, \\ \sqrt{x}+1, & 0\leq x<4, \\ \frac{x^2-1}{x-1}, & x\geq 4 \end{cases}$$

Que 4 Typeset the following matrix and label its different sections using :

$$\left[\begin{array}{ccccc} \overbrace{\quad\quad\quad}^{\text{Column Group 1}} & & & \overbrace{\quad\quad}^{\text{Column Group 2}} & \\ a_{11} & a_{12} & a_{13} & a_{14} & a_{15} \\ a_{21} & a_{22} & a_{23} & a_{24} & a_{25} \\ a_{31} & a_{32} & a_{33} & a_{34} & a_{35} \end{array}\right]$$

Que 7 Create the following table.

Academic Performance		
a	b	c
1	2	3
1	2	3

Que 8 Create the following table.

Department	Course	Credits	Grade
CSE	Data Structures	4	A+
	Machine Learning	4	A+
	Operating Systems	3	B+
ECE	Digital Electronics	4	A
	Signals	3	B+

Que 9 Create the following scientific table. **Que 9** Create the following scientific table.

Department	Course	Credits	Grade
	Data Structures	4	A+
	Machine Learning	4	A+
	Operating Systems	3	B+
ECE	Digital Electronics	4	A
	Signals	3	B+

Que 10 Create a student-performance table.

Student	Mathematics	Physics	Computer Science
Amit	92	81	95
Neha	72	88	91
Rahul	68	74	79
Priya	96	94	98